

Problem Set 4: Sequences, Series, and Integration

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This problem set covers convergence tests for infinite series, techniques of integration, and an introduction to linear systems via matrix notation. Show all work; partial credit is awarded for correct setup even when the final numeric answer is wrong.

Problem 1

Let $\{a_n\}_{n=1}^{\infty}$ be the sequence defined by $a_n = \frac{2n^2 + 3n - 1}{5n^2 + n}$.

- (a) Compute $\lim_{n \rightarrow \infty} a_n$, showing your reasoning.
- (b) Determine whether the series $\sum_{n=1}^{\infty} (a_n - \frac{2}{5})$ converges or diverges, and justify your answer using an appropriate test.
- (c) Is the sequence $\{a_n\}$ monotonic for $n \geq 1$? Prove your claim.

Solution. For part (a), divide numerator and denominator by n^2 :

$$a_n = \frac{2 + 3/n - 1/n^2}{5 + 1/n} \rightarrow \frac{2}{5} \quad \text{as } n \rightarrow \infty. \quad (1)$$

By (1), $\lim_{n \rightarrow \infty} a_n = \frac{2}{5}$, so $a_n - \frac{2}{5} \rightarrow 0$, which is only a necessary (not sufficient) condition for convergence of the associated series. A more careful expansion gives

$$a_n - \frac{2}{5} = \frac{5(2n^2 + 3n - 1) - 2(5n^2 + n)}{5n(5n + 1)} = \frac{13n - 5}{5n(5n + 1)},$$

which behaves like $\frac{13}{25n}$ for large n , so by the limit comparison test with the harmonic series $\sum 1/n$, the series in part (b) *diverges*.

Problem 2

Evaluate the following integrals.

(i) $\int_0^{\pi/2} \sin^3(x) \cos(x) dx$

- (ii) $\int \sqrt{4-x^2} dx$ using a trigonometric substitution
- (iii) $\int_1^\infty \frac{1}{x^2+1} dx$, identifying it as an improper integral

Solution. For (i), let $u = \sin x$, so $du = \cos x dx$ and the bounds become $u = 0$ to $u = 1$:

$$\begin{aligned} \int_0^{\pi/2} \sin^3(x) \cos(x) dx &= \int_0^1 u^3 du \\ &= \left[\frac{u^4}{4} \right]_0^1 \\ &= \frac{1}{4}. \end{aligned} \tag{2}$$

For (ii), substitute $x = 2 \sin \theta$, $dx = 2 \cos \theta d\theta$, so that $\sqrt{4-x^2} = 2 \cos \theta$ and

$$\int \sqrt{4-x^2} dx = \int 4 \cos^2 \theta d\theta = 2\theta + \sin(2\theta) + C = 2 \arcsin\left(\frac{x}{2}\right) + \frac{x\sqrt{4-x^2}}{2} + C.$$

For (iii), we recognise the antiderivative $\arctan x$ and take a limit:

$$\begin{aligned} \int_1^\infty \frac{1}{x^2+1} dx &= \lim_{b \rightarrow \infty} \left[\arctan x \right]_1^b \\ &= \lim_{b \rightarrow \infty} \arctan b - \arctan 1 \\ &= \frac{\pi}{2} - \frac{\pi}{4} \\ &= \frac{\pi}{4}. \end{aligned}$$

This integral converges to $\pi/4$, since $\arctan b$ is bounded as $b \rightarrow \infty$.

Problem 3

Consider the linear system

$$\begin{cases} 2x + y - z = 4 \\ x - 3y + 2z = -6 \\ 3x + 2y + z = 5 \end{cases}$$

which can be written in matrix form $A\mathbf{x} = \mathbf{b}$ with

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & -3 & 2 \\ 3 & 2 & 1 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 4 \\ -6 \\ 5 \end{pmatrix}.$$

- (a) Compute $\det(A)$ and determine whether A is invertible.
- (b) If A is invertible, describe (without full computation) how Cramer's rule would be used to solve for z .

Solution. Expanding along the first row,

$$\det(A) = 2[(-3)(1) - (2)(2)] - 1[(1)(1) - (2)(3)] + (-1)[(1)(2) - (-3)(3)].$$

Simplifying each bracket,

$$\det(A) = 2(-3 - 4) - 1(1 - 6) - 1(2 + 9) = 2(-7) - 1(-5) - 1(11) = -14 + 5 - 11 = -20.$$

Since $\det(A) = -20 \neq 0$, the matrix A is invertible, and the system has a unique solution $\mathbf{x} = A^{-1}\mathbf{b}$.

For part (b), Cramer's rule states that

$$z = \frac{\det(A_z)}{\det(A)},$$

where A_z is formed by replacing the third column of A with $\mathbf{b}$.

Problem 4

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = x^3 - 3x + 1$.

- (a) Find all critical points of f and classify each as a local maximum, local minimum, or neither.
- (b) Sketch the general shape of the graph, noting concavity changes.

Solution. We have $f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1)$, so the critical points are $x = 1$ and $x = -1$. Since $f''(x) = 6x$, we get $f''(1) = 6 > 0$ (local minimum) and $f''(-1) = -6 < 0$ (local maximum). The inflection point occurs where $f''(x) = 0$, i.e. at $x = 0$, where concavity switches from concave-down (for $x < 0$) to concave-up (for $x > 0$). A useful sanity check on the algebra above is the bound

$$|f(x) - f(1)| \leq |x^3 - 3x| + 2 \quad \text{for } x \in [-2, 2],$$

which follows directly from the triangle inequality applied term by term.

It is also worth recording the values of f and f' at the critical points side by side, since this pair of columns is exactly what we will plot in the next section:

$$\begin{array}{ll} f(-1) = 3, & f'(-1) = 0, \\ f(1) = -1, & f'(1) = 0. \end{array}$$

Finally, the closed-form expression for the local extrema in terms of a generic cubic $g(x) = x^3 + px + q$ can be written compactly as

$$x_{\pm} = \left(\pm \sqrt{\frac{-p}{3}} \right),$$

which for $p = -3$ recovers $x_{\pm} = \pm 1$ as found above.